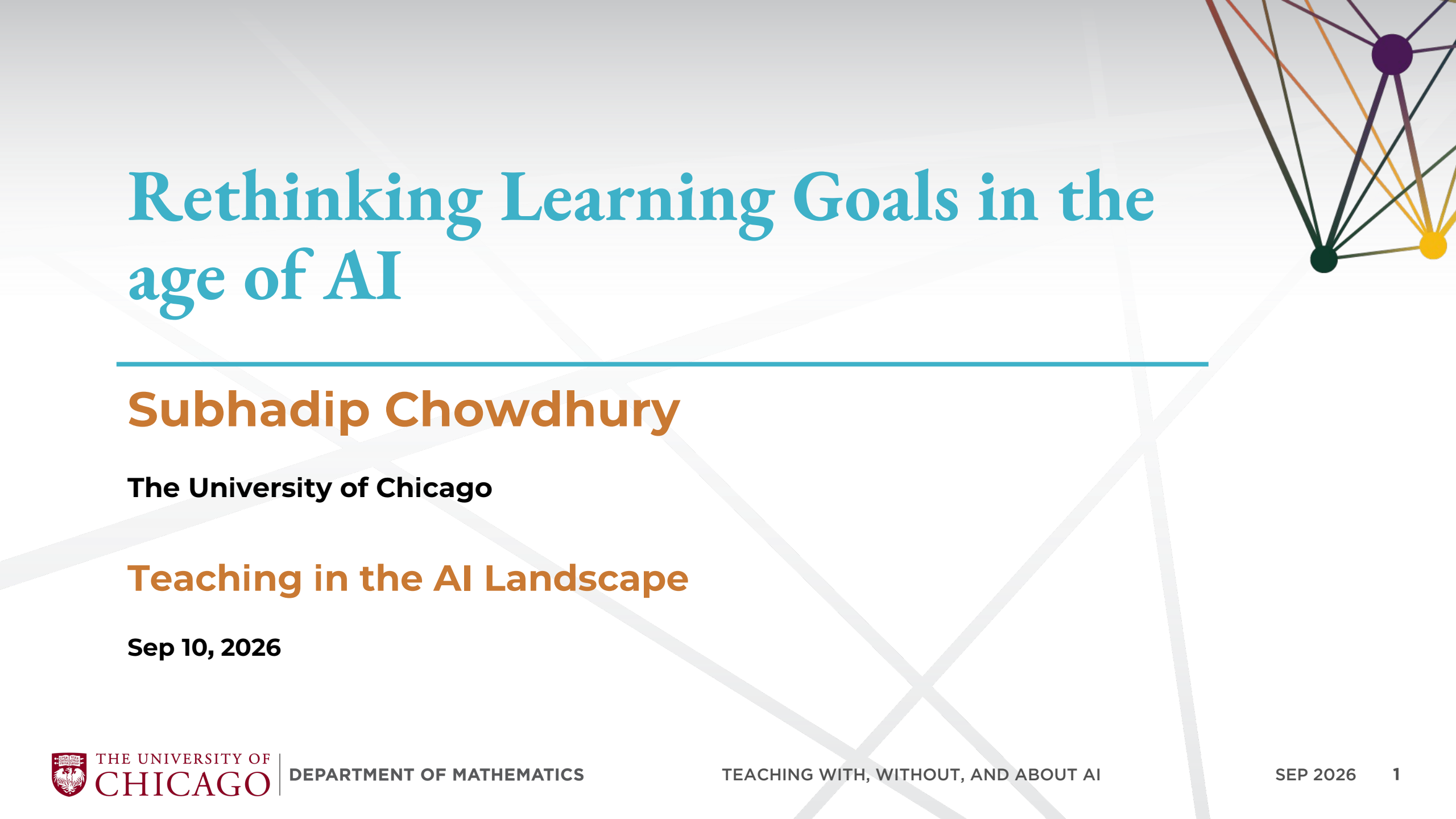


Rethinking Learning Goals in the age of AI



Subhadip Chowdhury

The University of Chicago

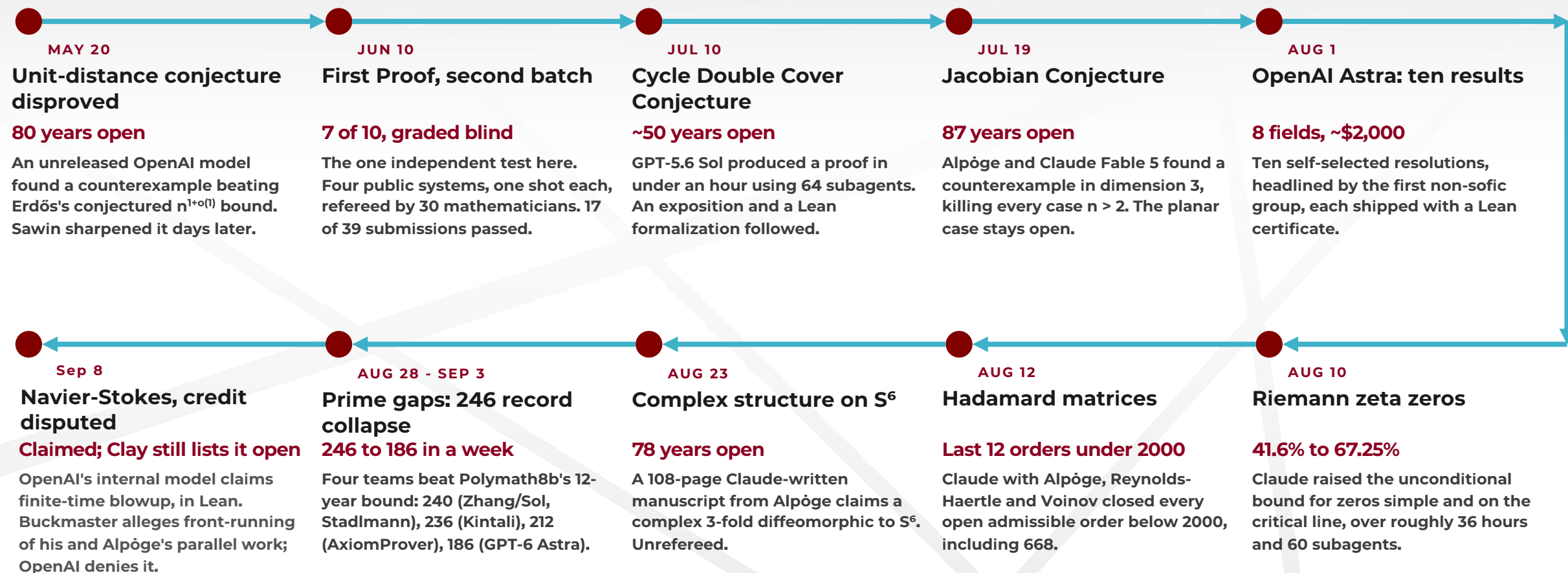
Teaching in the AI Landscape

Sep 10, 2026



Major results in AI and mathematics research

Ten milestones, May to September 2026. Dates mark public announcement.



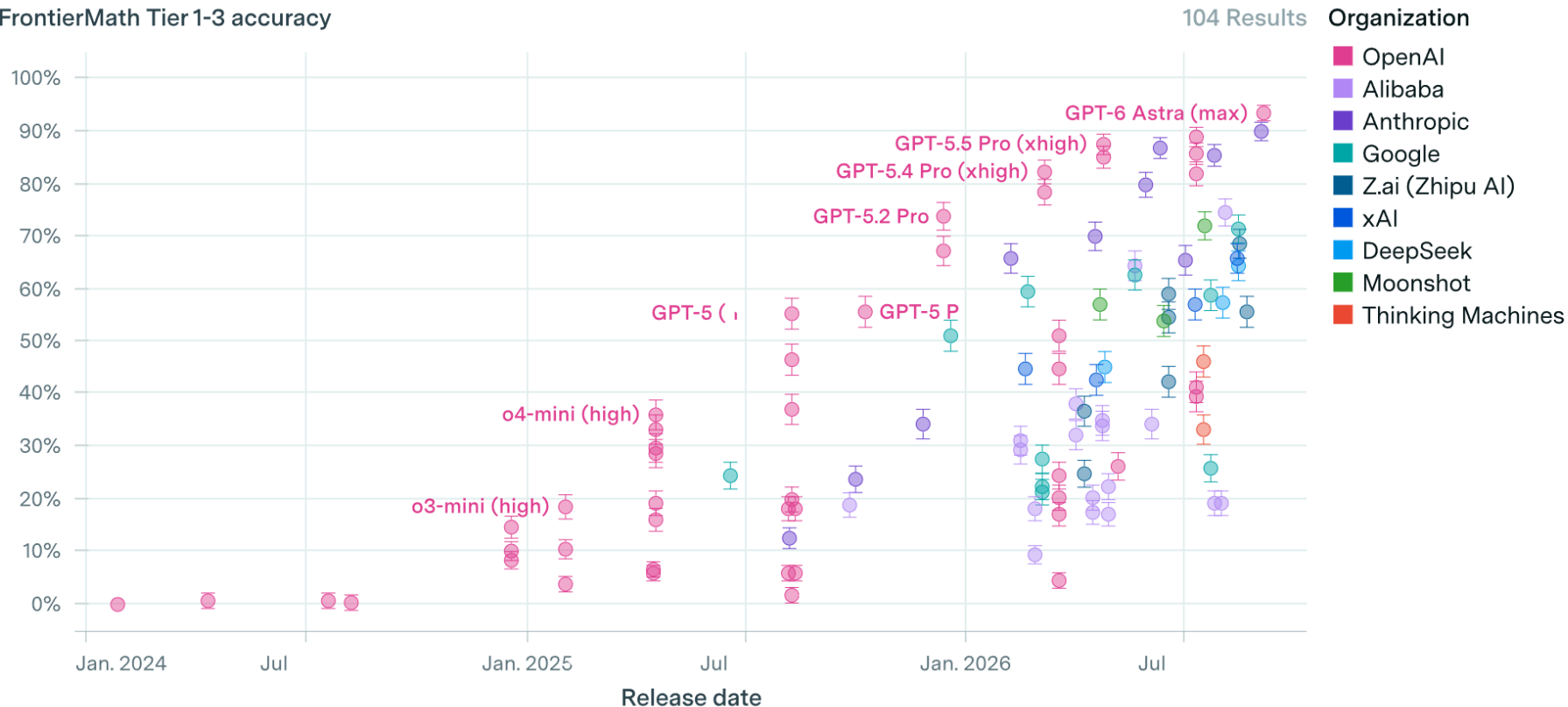
This slide was drafted with AI assistance. All sources were checked against primary documents on 7 September 2026.



AI in Undergraduate Math

AI performance on a set of expert-level mathematics problems

FrontierMath Tier 1-3 accuracy



EPOCH AI | CC-BY

epoch.ai

Retrieved 7 September 2026.



THE UNIVERSITY OF
CHICAGO

DEPARTMENT OF MATHEMATICS

TEACHING WITH, WITHOUT, AND ABOUT AI

SEP 2026

3



Measure vs. Goal

Adapted from Terence Tao, "Mathematics in the age of AI," <https://arxiv.org/pdf/2608.16753>

WORKING HYPOTHESIS

AI tools will, *reasonably* soon, be able to complete a *reasonable* fraction of the work we assign in our courses — with a *reasonable* level of success, quality, cost, and student initiative.

You don't have to believe this. Just assume it's true for the rest of the talk. How will we respond?

The measure (what we grade)

- Correct homework set
- Well-written proofs
- Completed the syllabus content

The goal (what we value)

- Student persistence through confusion
- Student gained the judgment to tell whether their own answer is reasonable
- Student is ready for the next course

These used to be positively correlated. But excessive AI optimization made them diverge.

Goodhart's law (1975): when a measure becomes a target, it ceases to be a good measure.





The Goal, Restated

Adapted from Terence Tao, "Mathematics in the age of AI," <https://arxiv.org/pdf/2608.16753>

Students produce correct work.

AI tools already do this at the coursework level. This layer is the one most exposed to the working hypothesis.

...and they can verify it themselves.

Students are starting to submit correct work that reads correctly to an AI, but that they cannot reconstruct themselves.

...and they can explain it to someone else.

Exposition is a poor optimization target. AI output is often too clean, strangely weighted, and blind to connections with prior knowledge.

...and have it be digested and accepted by a peer.

The communication needs some 'natural friction' — signals that help the reader see which parts were difficult and where the author struggled.

...and it becomes part of their learning process.

No AI tool can accelerate this one.

Flawless work on an assignment used to be scarce, and therefore informative. It is now abundant, and therefore uninformative, while none of the downstream stages have moved at all.



How am I responding?

The later stages require genuine peer-to-peer interaction and can be supervised in a classroom setting.

1 Discovery and Inquiry in the classroom

Interactive worksheets and Collaborative Learning.

2 Codify my Pedagogical Content Knowledge

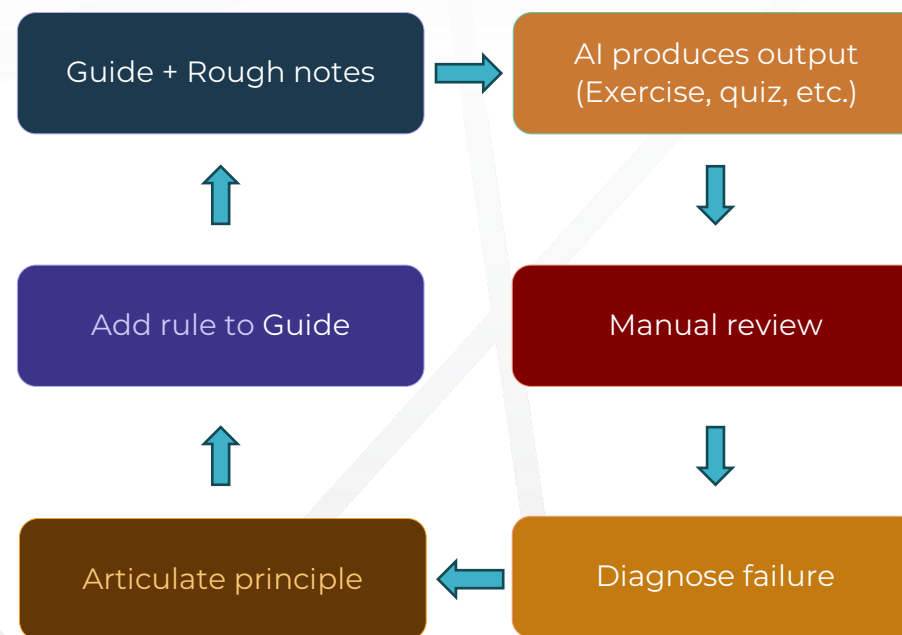
Use an AI model to help me produce lecture material at volume while saving time. [Teaching Your AI How You Teach](#)

3 Assignments rebuilt around AI

[\(Labs and materials\)](#)

- Numerical analysis Parsons problems lab.
- Concept maps.
- Simulations.

The Pedagogical Style Guide



Each cycle converts one piece of implicit PCK into an explicit rule.



Thank you for listening!

Questions?

subhadipchowdhury.github.io
subhadip@uchicago.edu

References

- Terence Tao, “Mathematics in the age of AI,” plenary lecture, International Congress of Mathematicians, 24 July 2026.
- Leiden Declaration on AI and Mathematics, endorsed by the International Mathematical Union. leidendeclaration.ai
- Lee Shulman, “Those who understand: knowledge growth in teaching,” *Educational Researcher* 15(2), 1986.
- Michael Polanyi, *The Tacit Dimension*, 1966.
- Subhadip Chowdhury, “Teaching Your AI How You Teach,” CCTL Teaching Matters, 1 September 2026.

AI DISCLOSURE Claude (Anthropic) was used for prose editing in this presentation.

